25901272 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFTQ201 - FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer type

Answer any **5** questions

Each question carries **2** marks

1. Where do colonies grow in pour plate? [U] / [CO1]
2. State one limitation of continuous culture systems. [K] / [CO3]
3. What toxin is produced by *Shigella dysenteriae*? [U] / [CO4]
4. Example of differential media. [U] / [CO1]
5. What is the effect of temperature on microbial growth? [K] / [CO3]
6. What is phenol used for? [E] / [CO1]
7. Define compound microscope. [U] / [CO1]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Outline the laboratory methods used for isolation and identification of Listeria monocytogenes from food and clinical samples. [K] / [CO4]

9. Classify bacteria based on their shape and arrangement with examples. [U] / [CO3]

10. Working of fluorescent microscope. [U] / [CO1]

11. Outline the pathogenesis of Bacillus cereus food intoxication. [U] / [CO1]

12. Explain the significance of bacterial arrangements in identifying bacterial species. [U] / [CO3]

13. Explain the pathogenesis of enterohemorrhagic E. coli (EHEC) infection. [K] / [CO4]

14. Explain Koch's postulates with applications. [U] / [CO1]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Compare and contrast staphylococcal food intoxication with food infections caused by Salmonella and Clostridium perfringens. [K] / [CO1]

16. Detailed comparison of TEM and SEM. [U] / [CO1, CO2]

17. Explain the prevention and control measures of Clostridium perfringens food infection in food handling and storage. [An] / [CO4]

18. Explain the prevention and control measures of salmonellosis at household and community levels. [U] / [CO4]

[10x2 = 20]